

NIST SP 800-22 Rev. 1a Evaluation of the Authenticated-TPE Method

Scope and interpretation

This report presents separate statistical diagnostics for authenticated-TPE cryptographic components and for the image outputs. NIST SP 800-22 is a statistical test suite, not a security proof. A test outcome does not establish security or insecurity by itself.

Method and implementation configuration

The implementation processes 512×512 RGB uint8 arrays in non-overlapping 32×32 blocks. It uses 256 blocks per channel, 512 adjacent pixel pairs per block, groups four ordered blocks, 64 groups per channel and 192 groups per RGB image. Authentication uses HMAC-SHA256; grouping uses the implemented HMAC-DRBG; transforms are Step 1 pair-sum-preserving mapping, Step 2 image-derived keyed smaller-pixel re-encryption, and reversible contrast mapping (RCM) for tag embedding. Verification and exact recovery occur before plaintext release.

The cryptographic-component categories test outputs used by the implementation. The Authentication-tag stream concatenates raw 256-bit HMAC-SHA256 group tags produced under independent test keys and ImageIDs; an individual 256-bit tag is not represented as a 1,000,000-bit stream. The image-output categories are explicitly diagnostic and preserve the method's image-dependent structure.

NIST software and settings

- Software: official NIST Statistical Test Suite 2.1.2, selected from the NIST SP 800-22 documentation and software page.
- Tests: all 15 STS tests; alpha = 0.01; default parameters (block frequency 128, non-overlapping templates m=9, overlapping template m=9, approximate entropy m=10, serial m=16, linear complexity M=500).
- Sample: 100 streams for each category; each stream is exactly 1,000,000 bits; 600 streams total across six categories.
- Input: NIST binary mode. The runner confirmed 100 BITSREAD = 1000000 records per category. Each 125,000-byte stream is consumed MSB-first, matching the NIST convertToBits implementation.
- This report uses a precision-only rebuild of official NIST STS 2.1.2. NIST test algorithms, formulas, thresholds, alpha, parameters, and input streams are unchanged. Floating-point report formatting uses 17 significant digits, and assess.c reads serialized values as double so second-level calculations use full computed precision. The exact patch and SHA-256 are included.
- Original reports from the unmodified STS binary are preserved under official_native_6dp/; the original rounded values are not used as the source for high-precision CSV values.
- Deterministic component fixture bytes use SHA-256(domain_label || 0x00 || uint64_be(index)), truncated only where a 16-byte ImageID is needed. The exact domain labels, per-stream key/nonce digests, and fixture indexes are recorded in stream_manifest.csv and component_contexts.csv.

Exact stream-generation procedure

- cryptographic_hmac_drbg: 100 independent streams. Each uses a unique 32-byte deterministic test entropy input and 16-byte nonce, then the authenticated-TPE HMAC_DRBG(SHA-256) implementation with personalization 'group-perm'. Each stream concatenates 31,250 successive 4-byte Generate outputs (1,000,000 bits). This extends the same implementation beyond the 255 Fisher-Yates draws typically consumed per channel; no value is repeated or padded.
- cryptographic_pair_shift_words: 100 independent UserKey/ImageID fixtures. Each stream contains the first 31,250 distinct valid (channel, block_id, pair_index) HMAC inputs, ordered channel 0..2, block 0..255, pair 0..511. Each output is the exact first 32 digest bits used by pair_shift before the implementation's class-size modulo. This tests the raw HMAC-derived word, not a fabricated image and not post-modulo shift values.
- cryptographic_block_r1_values: Each 1,000,000-bit sequence is the exact two-bit block_r1 output across independent UserKey contexts in context/channel/block order. A single context yields only 768 values, so 652 independent key contexts are used per NIST stream; the last context contributes only the 32 values needed for the exact prefix. Each r1 is encoded as two MSB-first bits. Keys are not reused and no values are repeated or padded.

- Authentication-tag stream: 100 separate NIST bitstreams are assembled from raw 256-bit HMAC-SHA256 group tags generated by the implementation `_group_mac`. Each stream uses 21 independent UserKey/ImageID contexts; all 2,100 pairs are unique. Six fixed Step-1 public UCT reference images supply authenticated group data round-robin. Tags are concatenated in context/channel/group order (4,032 tags generated per sequence) and truncated at exactly 1,000,000 bits, within the final tag bytes. This is an isolated tag-component diagnostic, not 100 full image-protection runs.
- `image_output_step2`: the first 1,000,000 bits of row-major C-order RGB uint8 bytes from the Step-2 image-derived keyed intermediate, before authentication-tag embedding.
- `image_output_final_marked`: the first 1,000,000 bits of row-major C-order RGB uint8 bytes from the final RCM-marked authenticated image.

Image and key fixture selection

The image diagnostic uses the six available public UCT images in round-robin order: airplane, baboon, couple, girl, lena, peppers. Images are converted to RGB and resized to 512×512 only when required, using Pillow's default resize behavior to match the existing runner. For stream `i`, UserKey is `SHA-256(auth-tpe-nist-key-v1 || 00 || uint32_be(i))` and ImageID is the first 16 bytes of `SHA-256(auth-tpe-nist-image-id-v1 || 00 || uint32_be(i))`. The paired Step-2 and marked outputs for one stream use the same test credentials. These are reproducibility fixtures, not production secrets. The exact eight CelebA-HQ source files were unavailable; no CelebA-HQ data is claimed or fabricated.

The four cryptographic-component fixture families use domain-separated deterministic public bytes: HMAC-DRBG entropy/nonce, pair-shift key/ImageID, block-r1 keys, and authentication-tag key/ImageID. The tag category uses 2,100 unique UserKey/ImageID pairs across the 100 streams. These fixtures support repeatability and are public; they do not test entropy sources or secret-key security.

Byte-to-bit serialization

Cryptographic byte outputs are recorded in generation order. For the image categories, the tested object is row-major C-order RGB uint8 bytes. The first 125,000 bytes are the exact 1,000,000-bit prefix. All inputs are passed as packed binary; NIST reads every byte from its most significant bit to its least significant bit. No LSB-first or full-image variant was run, and no image-output padding was used.

Results overview

Cryptographic-component evaluation

Category	Pass	Fail	N/A
cryptographic_hmac_drbg	17328	172	1300
cryptographic_pair_shift_words	17537	171	1092
cryptographic_block_r1_values	17698	192	910
Authentication-tag stream	17631	181	988

Image-output statistical diagnostic

Category	Pass	Fail	N/A	Exact recovery
Encrypted Step-2 image-output diagnostic	1838	14362	2600	100/100
Final encrypted image with embedded authentication information	1938	14262	2600	100/100

Step 1 preserves pair sums and therefore retains source-image structure in the encrypted intermediate. Reversible contrast mapping (RCM) embeds authentication information reversibly in the final marked image. Both image-output categories retain image-dependent structure and are statistical diagnostics; they are not expected to behave like ideal random streams. Low p-values in these categories do not by themselves establish an implementation defect. The HMAC/DRBG component diagnostics are more appropriate for pseudorandom behavior, but NIST SP 800-22 does not prove security.

A 0 from the precision-patched STS is not six-decimal rounding. For Runs, an algorithm-defined zero indicates the NIST PI estimator criterion was not met. For other applicable tests, binary64 returned zero; the CSV labels this `double_underflow` and gives the conservative bound $p < 1e-300$. These differ from N/A, which has a blank p-value and is excluded from valid-stream counts.

The very small Frequency p-values in image outputs follow from the observed NIST signed partial sums S_n : $|S_n|$ spans 12,046 to 208,006 bits (median 94,214.0) for the Encrypted Step-2 image-output diagnostic, and 13,996 to 207,700 bits (median 100,749.0) for the final marked image. For $N=1,000,000$, the Frequency p-value is $\text{erfc}(|S_n|/\sqrt{2N})$; these large deviations from a balanced bit count yield extremely small tail probabilities, with the largest values below binary64 range. This is consistent with retained image structure. The full S_n , p-values, and other test statistics are preserved in the per-stream STS stats files.

cryptographic_hmac_drbg

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	99	1	0	0.30412581641191033	99/100
BlockFrequency	single	100	99	1	0	0.15376333779556456	99/100
CumulativeSums	forward	100	99	1	0	0.051941639795524047	99/100
CumulativeSums	reverse	100	99	1	0	0.28966749038847861	99/100
Runs	single	100	98	2	0	0.18155663198089642	98/100
LongestRun	single	100	97	3	0	0.97169915710286503	97/100
Rank	single	100	100	0	0	0.88317137842416837	100/100
FFT	single	100	100	0	0	0.22482064155775411	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	100	0	0	0.21330930508341639	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	100	0	0	0.99424994542029488	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	99	1	0	0.53414621690969133	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	98	2	0	0.31908350214261383	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	100	0	0	0.045674574411820157	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	97	3	0	0.98789554017452808	97/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	99	1	0	0.96429497268508912	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	100	0	0	0.14532611574989807	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	99	1	0	0.15376333779556461	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	97	3	0	0.23680981521773165	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	99	1	0	0.59554850728420583	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	100	0	0	0.38382651737138401	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	99	1	0	0.92407593012659839	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	100	0	0	0.36691779911275224	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	99	1	0	0.73991829209465365	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	98	2	0	0.67868578529607482	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	99	1	0	0.53414621690969155	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	99	1	0	0.055361051941826951	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	99	1	0	0.98789554017452808	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	100	0	0	0.042808281757667772	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	100	0	0	0.67868578529607482	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	98	2	0	0.5141236202310755	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	100	0	0	0.289667490388479	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	98	2	0	0.12961959133276191	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	98	2	0	0.26224875459496472	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	98	2	0	0.49439168648212251	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0	100	99	1	0	0.8977625971214902	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	99	1	0	0.88317137842416837	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 0	100	100	0	0	0.24928389397664924	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	100	0	0	0.73991829209465365	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	99	1	0	0.9914676066288135	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	100	0	0	0.21330930508341672	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	99	1	0	0.1153865825888612	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	100	0	0	0.31908350214261383	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	99	1	0	0.85138257535679496	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	98	2	0	0.77918771616483629	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	99	1	0	0.75975627887206953	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	100	0	0	0.45593719522066189	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	100	0	0	0.22482064155775411	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	98	2	0	0.6993125708664083	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0	100	99	1	0	0.1153865825888612	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1	100	98	2	0	0.63711940716939852	98/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0	100	97	3	0	0.6163052249833656	97/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1	100	98	2	0	0.289667490388479	98/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	99	1	0	0.5141236202310755	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	100	0	0	0.040108279765939552	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	100	0	0	0.058984030544419419	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	100	0	0	0.79813906239536136	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	100	0	0	0.6993125708664083	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	99	1	0	0.86769167149770521	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	100	0	0	0.86769167149770521	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	100	0	0	0.24928389397664924	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	99	1	0	0.19168674084696566	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	100	0	0	0.65793330621362878	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	100	0	0	0.55442043587285705	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	100	0	0	0.2757089367722202	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 0 1 1	100	99	1	0	0.67868578529607482	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	100	0	0	0.43727418891386649	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 0 1 1	100	98	2	0	0.12232522803866246	98/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	99	1	0	0.43727418891386738	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 0 1 1	100	100	0	0	0.048715976147641296	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	100	0	0	0.8165367975972474	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	97	3	0	0.8977625971214902	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	100	0	0	0.73991829209465376	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	100	0	0	0.85138257535679507	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	100	0	0	0.31908350214261433	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	98	2	0	0.33453815163400613	98/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	99	1	0	0.95583472560026084	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	97	3	0	0.33453815163400619	97/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	99	1	0	0.8977625971214902	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	100	0	0	0.85138257535679496	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	99	1	0	0.49439168648212273	99/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	99	1	0	0.24928389397664924	99/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	100	0	0	0.77918771616483629	100/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	100	0	0	0.21330930508341639	100/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	99	1	0	0.73991829209465365	99/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	97	3	0	0.30412581641191033	97/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	98	2	0	0.53414621690969155	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	98	2	0	0.97169915710286503	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	100	0	0	0.55442043587285705	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	98	2	0	0.53414621690969155	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 1 0 0	100	100	0	0	0.47498568664800966	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	98	2	0	0.289667490388479	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	98	2	0	0.23680981521773165	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	99	1	0	0.43727418891386649	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	99	1	0	0.73991829209465365	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	99	1	0	0.2022676854058254	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	98	2	0	0.67868578529607482	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	99	1	0	0.33453815163400613	99/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	100	0	0	0.22482064155775419	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	100	0	0	0.31908350214261388	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	100	0	0	0.67868578529607482	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	100	0	0	0.15376333779556461	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	100	0	0	0.1025256802140477	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	96	4	0	0.67868578529607482	96/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	97	3	0	0.16260626219029947	97/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	100	0	0	0.5141236202310755	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 1 0	100	99	1	0	0.97169915710286503	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	99	1	0	0.36691779911275224	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	100	0	0	0.12232522803866246	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	100	0	0	0.24928389397664924	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	99	1	0	0.45593719522066167	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	100	0	0	0.88317137842416837	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	98	2	0	0.92407593012659839	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	99	1	0	0.41902116661019273	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	100	0	0	0.41902116661019245	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	100	0	0	0.49439168648212251	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	97	3	0	0.55442043587285705	97/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	99	1	0	0.23680981521773165	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	99	1	0	0.075719263168633866	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	98	2	0	0.080518512818763724	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	98	2	0	0.40119928658025733	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	99	1	0	0.023545032770843879	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	98	2	0	0.042808281757667772	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	100	0	0	0.035173539466984836	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	99	1	0	0.01918759798300609	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	99	1	0	0.61630522498336537	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	99	1	0	0.33453815163400613	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	100	0	0	0.051941639795524061	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	99	1	0	0.33453815163400619	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	99	1	0	0.47498568664800966	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	99	1	0	0.35048521232336116	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	100	0	0	0.91141252683167917	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	98	2	0	0.47498568664800989	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	98	2	0	0.31908350214261383	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	99	1	0	0.040108279765939524	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	100	0	0	0.16260626219029947	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	98	2	0	0.096578173851826141	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	99	1	0	0.71974657482403415	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	100	0	0	0.080518512818763724	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	99	1	0	0.28966749038847928	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	97	3	0	0.2757089367722218	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	99	1	0	0.0095349098257501777	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	98	2	0	0.92407593012659839	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	99	1	0	0.066881587774126644	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	100	0	0	0.058984030544419586	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	98	2	0	0.10252568021404759	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	100	0	0	0.5141236202310755	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	98	2	0	0.99633468910414835	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	98	2	0	0.1153865825888612	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	98	2	0	0.83430826019340754	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	99	1	0	0.59554850728420583	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	98	2	0	0.65793330621362878	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 1 0	100	100	0	0	0.00047444911427182404	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0 0	100	100	0	0	0.38382651737138435	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 1 0	100	99	1	0	0.73991829209465365	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0 0	100	99	1	0	0.090935976579805164	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 1 0	100	100	0	0	0.81653679759724718	100/100
OverlappingTemplate	single	100	96	4	0	0.91141252683167917	96/100
Universal	single	100	97	3	0	0.5141236202310755	97/100
ApproximateEntropy	single	100	99	1	0	0.12961959133276191	99/100
RandomExcursions	state=-4	50	50	0	50	0.6993125708664083	50/50

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
RandomExcursions	state=-3	50	49	1	50	0.93571641091397317	49/50
RandomExcursions	state=-2	50	50	0	50	0.26224875459496477	50/50
RandomExcursions	state=-1	50	50	0	50	0.97169915710286503	50/50
RandomExcursions	state=1	50	50	0	50	0.13728235312658429	50/50
RandomExcursions	state=2	50	50	0	50	0.38382651737138435	50/50
RandomExcursions	state=3	50	50	0	50	0.49439168648212251	50/50
RandomExcursions	state=4	50	50	0	50	0.096578173851826141	50/50
RandomExcursionsVariant	state=-9	50	49	1	50	0.23680981521773181	49/50
RandomExcursionsVariant	state=-8	50	49	1	50	0.10879095519400765	49/50
RandomExcursionsVariant	state=-7	50	49	1	50	0.013568668659574932	49/50
RandomExcursionsVariant	state=-6	50	49	1	50	0.35048521232336116	49/50
RandomExcursionsVariant	state=-5	50	49	1	50	0.289667490388479	49/50
RandomExcursionsVariant	state=-4	50	49	1	50	0.49439168648212251	49/50
RandomExcursionsVariant	state=-3	50	49	1	50	0.6163052249833656	49/50
RandomExcursionsVariant	state=-2	50	50	0	50	0.6993125708664083	50/50
RandomExcursionsVariant	state=-1	50	49	1	50	0.23680981521773165	49/50
RandomExcursionsVariant	state=1	50	49	1	50	0.49439168648212251	49/50
RandomExcursionsVariant	state=2	50	50	0	50	0.26224875459496477	50/50
RandomExcursionsVariant	state=3	50	50	0	50	0.6993125708664083	50/50
RandomExcursionsVariant	state=4	50	50	0	50	0.21330930508341639	50/50
RandomExcursionsVariant	state=5	50	50	0	50	0.38382651737138435	50/50
RandomExcursionsVariant	state=6	50	50	0	50	0.19168674084696535	50/50
RandomExcursionsVariant	state=7	50	50	0	50	0.6993125708664083	50/50
RandomExcursionsVariant	state=8	50	50	0	50	0.49439168648212273	50/50
RandomExcursionsVariant	state=9	50	50	0	50	0.15376333779556461	50/50
Serial	p_value1 (delta_1)	100	100	0	0	0.22482064155775411	100/100
Serial	p_value2 (delta_2)	100	100	0	0	0.080518512818763793	100/100
LinearComplexity	single	100	99	1	0	0.40119928658025733	99/100

cryptographic_pair_shift_words

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	99	1	0	0.19168674084696558	99/100
BlockFrequency	single	100	98	2	0	0.19168674084696535	98/100
CumulativeSums	forward	100	98	2	0	0.85138257535679496	98/100
CumulativeSums	reverse	100	98	2	0	0.94630767376442704	98/100
Runs	single	100	99	1	0	0.88317137842416837	99/100
LongestRun	single	100	99	1	0	0.24928389397664924	99/100
Rank	single	100	100	0	0	0.5141236202310755	100/100
FFT	single	100	99	1	0	0.12232522803866246	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	98	2	0	0.49439168648212251	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	99	1	0	0.57490342386445548	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	99	1	0	0.53414621690969155	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	99	1	0	0.23680981521773181	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	99	1	0	0.73991829209465365	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	99	1	0	0.63711940716939863	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	98	2	0	0.79813906239536148	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	98	2	0	0.13728235312658429	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	97	3	0	0.49439168648212251	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	98	2	0	0.28966749038847928	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	98	2	0	0.071177409391062527	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	100	0	0	0.43727418891386738	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	100	0	0	0.91141252683167917	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	99	1	0	0.055361051941826937	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	99	1	0	0.33453815163400613	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	97	3	0	0.79813906239536136	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	100	0	0	0.16260626219029947	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	100	0	0	0.16260626219029947	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	97	3	0	0.61630522498336548	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0 1	100	99	1	0	0.31908350214261383	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1 1	100	98	2	0	0.26224875459496472	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0 1	100	98	2	0	0.41902116661019245	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1 1	100	99	1	0	0.6993125708664083	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1 1	100	100	0	0	0.20226768540582532	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0 1	100	100	0	0	0.65793330621362878	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1 1	100	99	1	0	0.71974657482403437	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0 1	100	97	3	0	0.45593719522066167	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1 1	100	99	1	0	0.65793330621362878	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 0 1	100	100	0	0	0.71974657482403392	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1 1	100	96	4	0	0.43727418891386738	96/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1 1	100	98	2	0	0.02199861694582168	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0 1	100	98	2	0	0.9780720677116449	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1 1	100	99	1	0	0.30412581641191033	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1 1	100	98	2	0	0.67868578529607482	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0 1	100	98	2	0	0.1153865825888612	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1 1	100	100	0	0	0.88317137842416837	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1 1	100	99	1	0	0.014549902304087557	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0 1	100	99	1	0	0.35048521232336122	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1 1	100	99	1	0	0.83430826019340754	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1 1	100	100	0	0	0.36691779911275224	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0 1	100	99	1	0	0.24928389397664924	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1 1	100	99	1	0	0.40119928658025733	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0 1	100	99	1	0	0.33453815163400619	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1 1	100	100	0	0	0.090935976579805164	100/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	100	0	0	0.31908350214261433	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	100	0	0	0.95583472560026084	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	100	0	0	0.63711940716939852	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	97	3	0	0.27570893677222202	97/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	99	1	0	0.55442043587285705	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	100	0	0	0.65793330621362878	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	100	0	0	0.13728235312658446	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	99	1	0	0.97169915710286503	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	99	1	0	0.16260626219029947	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	100	0	0	0.85138257535679496	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 0 1 1	100	99	1	0	0.30412581641191022	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	98	2	0	0.88317137842416837	98/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 0 1 1	100	99	1	0	0.23680981521773165	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	99	1	0	0.91141252683167917	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 0 1 1	100	99	1	0	0.5749034238644557	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	99	1	0	0.071177409391062527	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 0 1 1	100	97	3	0	0.98345295122950693	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	99	1	0	0.27570893677222202	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	100	0	0	0.38382651737138435	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	97	3	0	0.75975627887206942	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	100	0	0	0.53414621690969155	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	99	1	0	0.49439168648212251	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	99	1	0	0.59554850728420616	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	100	0	0	0.85138257535679496	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	98	2	0	0.33453815163400613	98/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	100	0	0	0.4011992865802575	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	100	0	0	0.85138257535679507	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	100	0	0	0.33453815163400619	100/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	97	3	0	0.33453815163400619	97/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	98	2	0	0.23680981521773165	98/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	98	2	0	0.49439168648212251	98/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	98	2	0	0.38382651737138401	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	100	0	0	0.8977625971214902	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	100	0	0	0.2022676854058254	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	100	0	0	0.38382651737138435	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	97	3	0	0.51412362023107572	97/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	98	2	0	0.93571641091397317	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 1 0 0	100	97	3	0	0.096578173851826141	97/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	99	1	0	0.98789554017452808	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	98	2	0	0.98789554017452808	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	99	1	0	0.41902116661019245	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	99	1	0	0.77918771616483629	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	97	3	0	0.040108279765939524	97/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	100	0	0	0.49439168648212251	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	100	0	0	0.98789554017452808	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	100	0	0	0.14532611574989807	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	98	2	0	0.289667490388479	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	98	2	0	0.36691779911275224	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	99	1	0	0.79813906239536148	99/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	100	0	0	0.45593719522066167	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	99	1	0	0.5141236202310755	99/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	99	1	0	0.0076941199843233322	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	99	1	0	0.47498568664800966	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 1 0	100	100	0	0	0.88317137842416837	100/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	100	0	0	0.24928389397664924	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	100	0	0	0.45593719522066167	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	100	0	0	0.67868578529607482	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	100	0	0	0.91141252683167917	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	100	0	0	0.2133093050834165	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	100	0	0	0.40119928658025733	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	100	0	0	0.86769167149770532	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	98	2	0	0.03292282173775303	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	98	2	0	0.79813906239536136	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	99	1	0	0.36691779911275224	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	99	1	0	0.85138257535679496	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	99	1	0	0.92407593012659839	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	98	2	0	0.1153865825888612	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	99	1	0	0.63711940716939852	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	100	0	0	0.045674574411820157	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	100	0	0	0.36691779911275224	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	100	0	0	0.657933306213629	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	99	1	0	0.36691779911275224	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	99	1	0	0.49439168648212251	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	100	0	0	0.30412581641191022	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	98	2	0	0.8165367975972474	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	100	0	0	0.73991829209465365	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	100	0	0	0.40119928658025733	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	98	2	0	0.88317137842416837	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	98	2	0	0.71974657482403415	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	99	1	0	0.35048521232336122	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	98	2	0	0.49439168648212251	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	100	0	0	0.6993125708664083	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	99	1	0	0.53414621690969133	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	99	1	0	0.8977625971214902	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	98	2	0	0.43727418891386738	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	100	0	0	0.38382651737138401	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	99	1	0	0.5749034238644557	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	99	1	0	0.010988016145239976	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	99	1	0	0.062820720617879772	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	100	0	0	0.23680981521773181	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	100	0	0	0.35048521232336122	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	100	0	0	0.95583472560026084	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	99	1	0	0.035173539466984677	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	98	2	0	0.5749034238644557	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	99	1	0	0.61630522498336537	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	98	2	0	0.41902116661019245	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	100	0	0	0.65793330621362878	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	97	3	0	0.88317137842416837	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	99	1	0	0.096578173851826016	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 1 0	100	99	1	0	0.12232522803866246	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0 0	100	99	1	0	0.19168674084696558	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 1 0	100	99	1	0	0.49439168648212251	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0 0	100	99	1	0	0.47498568664800989	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 1 0	100	98	2	0	0.23680981521773165	98/100
OverlappingTemplate	single	100	100	0	0	0.14532611574989807	100/100
Universal	single	100	100	0	0	0.289667490388479	100/100
ApproximateEntropy	single	100	100	0	0	0.02054794773160187	100/100
RandomExcursions	state=-4	58	56	2	42	0.066881587774126644	56/58

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
RandomExcursions	state=-3	58	57	1	42	0.657933306213629	57/58
RandomExcursions	state=-2	58	58	0	42	0.0015093583490445334	58/58
RandomExcursions	state=-1	58	58	0	42	0.38382651737138401	58/58
RandomExcursions	state=1	58	58	0	42	0.01356866865957492	58/58
RandomExcursions	state=2	58	58	0	42	0.38382651737138401	58/58
RandomExcursions	state=3	58	58	0	42	0.23680981521773181	58/58
RandomExcursions	state=4	58	58	0	42	0.35048521232336122	58/58
RandomExcursionsVariant	state=-9	58	58	0	42	0.73991829209465365	58/58
RandomExcursionsVariant	state=-8	58	57	1	42	0.23680981521773181	57/58
RandomExcursionsVariant	state=-7	58	58	0	42	0.011791392630075934	58/58
RandomExcursionsVariant	state=-6	58	58	0	42	0.8165367975972474	58/58
RandomExcursionsVariant	state=-5	58	58	0	42	0.657933306213629	58/58
RandomExcursionsVariant	state=-4	58	58	0	42	0.53414621690969155	58/58
RandomExcursionsVariant	state=-3	58	58	0	42	0.23680981521773128	58/58
RandomExcursionsVariant	state=-2	58	58	0	42	0.41902116661019245	58/58
RandomExcursionsVariant	state=-1	58	58	0	42	0.045674574411820157	58/58
RandomExcursionsVariant	state=1	58	58	0	42	0.051941639795524061	58/58
RandomExcursionsVariant	state=2	58	58	0	42	0.289667490388479	58/58
RandomExcursionsVariant	state=3	58	58	0	42	0.075719263168633755	58/58
RandomExcursionsVariant	state=4	58	58	0	42	0.26224875459496472	58/58
RandomExcursionsVariant	state=5	58	58	0	42	0.030806427866728648	58/58
RandomExcursionsVariant	state=6	58	58	0	42	0.21330930508341639	58/58
RandomExcursionsVariant	state=7	58	58	0	42	0.6993125708664083	58/58
RandomExcursionsVariant	state=8	58	58	0	42	0.41902116661019273	58/58
RandomExcursionsVariant	state=9	58	58	0	42	0.21330930508341672	58/58
Serial	p_value1 (delta_1)	100	100	0	0	0.289667490388479	100/100
Serial	p_value2 (delta_2)	100	100	0	0	0.53414621690969155	100/100
LinearComplexity	single	100	100	0	0	0.10252568021404759	100/100

cryptographic_block_r1_values

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	99	1	0	0.57490342386445548	99/100
BlockFrequency	single	100	98	2	0	0.59554850728420583	98/100
CumulativeSums	forward	100	99	1	0	0.0057624324498999893	99/100
CumulativeSums	reverse	100	99	1	0	0.95583472560026084	99/100
Runs	single	100	99	1	0	0.23680981521773181	99/100
LongestRun	single	100	99	1	0	0.096578173851826141	99/100
Rank	single	100	98	2	0	0.79813906239536159	98/100
FFT	single	100	99	1	0	0.21330930508341672	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	99	1	0	0.7791877161648364	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	99	1	0	0.93571641091397317	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	100	0	0	0.94630767376442704	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	100	0	0	0.062820720617879772	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	98	2	0	0.2022676854058254	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	98	2	0	0.55442043587285705	98/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	100	0	0	0.83430826019340754	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	98	2	0	0.8977625971214902	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	99	1	0	0.55442043587285705	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	99	1	0	0.65793330621362922	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	100	0	0	0.83430826019340754	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	98	2	0	0.30412581641191022	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	99	1	0	0.59554850728420583	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	97	3	0	0.85138257535679507	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	98	2	0	0.5749034238644557	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	99	1	0	0.85138257535679507	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	99	1	0	0.88317137842416837	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	100	0	0	0.36691779911275224	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	100	0	0	0.55442043587285705	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	100	0	0	0.49439168648212273	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	99	1	0	0.49439168648212273	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	99	1	0	0.86769167149770532	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	98	2	0	0.86769167149770521	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	98	2	0	0.35048521232336116	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	100	0	0	0.28966749038847928	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	99	1	0	0.10879095519400765	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0	100	100	0	0	0.94630767376442704	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	98	2	0	0.96429497268508912	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 0	100	100	0	0	0.18155663198089653	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	100	0	0	0.97169915710286503	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	100	0	0	0.9780720677116449	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	96	4	0	0.7791877161648364	96/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	100	0	0	0.17186683746684381	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	99	1	0	0.048715976147641296	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	98	2	0	0.85138257535679507	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	99	1	0	0.14532611574989807	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	100	0	0	0.9780720677116449	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	98	2	0	0.65793330621362922	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	100	0	0	0.22482064155775411	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	98	2	0	0.41902116661019245	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0	100	99	1	0	0.55442043587285705	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1	100	99	1	0	0.92407593012659839	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0	100	99	1	0	0.33453815163400613	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1	100	100	0	0	0.40119928658025733	100/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	99	1	0	0.41902116661019245	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	100	0	0	0.10879095519400762	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	100	0	0	0.59554850728420583	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	100	0	0	0.85138257535679496	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	99	1	0	0.71974657482403437	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	98	2	0	0.26224875459496477	98/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	100	0	0	0.59554850728420583	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	97	3	0	0.91141252683167917	97/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	98	2	0	0.33453815163400619	98/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	99	1	0	0.35048521232336116	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 0 1 1	100	99	1	0	0.12232522803866246	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	100	0	0	0.01559810019012501	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 0 1 1	100	97	3	0	0.93571641091397317	97/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	99	1	0	0.63711940716939863	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 0 1 1	100	97	3	0	0.23680981521773181	97/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	97	3	0	0.75975627887206953	97/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 0 1 1	100	99	1	0	0.062820720617879772	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	100	0	0	0.79813906239536136	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	100	0	0	0.55442043587285705	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	99	1	0	0.19168674084696566	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	99	1	0	0.6993125708664083	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	98	2	0	0.41902116661019245	98/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	99	1	0	0.5749034238644557	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	98	2	0	0.61630522498336537	98/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	97	3	0	0.40119928658025733	97/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	98	2	0	0.75975627887206953	98/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	99	1	0	0.67868578529607482	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	100	0	0	0.35048521232336122	100/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	99	1	0	0.93571641091397317	99/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	99	1	0	0.657933306213629	99/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	99	1	0	0.7791877161648364	99/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	100	0	0	0.037566083542570732	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	98	2	0	0.55442043587285705	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	100	0	0	0.38382651737138401	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	100	0	0	0.38382651737138435	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	100	0	0	0.17186683746684386	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	99	1	0	0.24928389397664924	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 1 0 0	100	99	1	0	0.058984030544419586	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	96	4	0	0.81653679759724718	96/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	99	1	0	0.33453815163400619	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	99	1	0	0.38382651737138401	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	96	4	0	0.10879095519400762	96/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	100	0	0	0.090935976579805164	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	98	2	0	0.23680981521773165	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	100	0	0	0.31908350214261383	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	100	0	0	0.59554850728420583	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	97	3	0	0.71974657482403437	97/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	99	1	0	0.062820720617879772	99/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	99	1	0	0.36691779911275224	99/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	98	2	0	0.55442043587285705	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	100	0	0	0.96429497268508912	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	100	0	0	0.35048521232336116	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	100	0	0	0.97169915710286503	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 1 0	100	99	1	0	0.63711940716939852	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	99	1	0	0.53414621690969155	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	99	1	0	0.6163052249833656	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	99	1	0	0.92407593012659839	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	98	2	0	0.36691779911275224	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	98	2	0	0.38382651737138401	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	97	3	0	0.35048521232336166	97/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	100	0	0	0.6163052249833656	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	100	0	0	0.45593719522066167	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	100	0	0	0.57490342386445548	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	99	1	0	0.71974657482403415	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	99	1	0	0.21330930508341639	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	97	3	0	0.75975627887206942	97/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	100	0	0	0.24928389397664924	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	100	0	0	0.4011992865802575	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	99	1	0	0.75975627887206942	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	98	2	0	0.23680981521773165	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	100	0	0	0.73991829209465365	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	99	1	0	0.75975627887206965	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	97	3	0	0.75975627887206942	97/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	99	1	0	0.014549902304087594	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	98	2	0	0.79813906239536136	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	99	1	0	0.94630767376442704	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	99	1	0	0.92407593012659839	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	99	1	0	0.47498568664800989	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	95	5	0	0.43727418891386738	95/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	98	2	0	0.20226768540582532	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	99	1	0	0.65793330621362878	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	100	0	0	0.73991829209465365	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	100	0	0	0.31908350214261388	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	97	3	0	0.83430826019340754	97/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	99	1	0	0.53414621690969155	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	99	1	0	0.92407593012659839	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	99	1	0	0.43727418891386738	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	98	2	0	0.31908350214261383	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	97	3	0	0.47498568664800989	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	98	2	0	0.31908350214261383	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	100	0	0	0.5141236202310755	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	99	1	0	0.31908350214261383	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	100	0	0	0.40119928658025733	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	100	0	0	0.91141252683167917	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	100	0	0	0.99977659451916134	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	100	0	0	0.65793330621362878	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	99	1	0	0.43727418891386738	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	100	0	0	0.23680981521773181	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	99	1	0	0.65793330621362878	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 1 0	100	100	0	0	0.4011992865802575	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0 0	100	99	1	0	0.040108279765939454	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 1 0	100	100	0	0	0.23680981521773165	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0 0	100	98	2	0	0.36691779911275224	98/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 1 0	100	99	1	0	0.71974657482403415	99/100
OverlappingTemplate	single	100	100	0	0	0.7791877161648364	100/100
Universal	single	100	100	0	0	0.55442043587285705	100/100
ApproximateEntropy	single	100	99	1	0	0.75975627887206942	99/100
RandomExcursions	state=-4	65	65	0	35	0.20407570932575123	65/65

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
RandomExcursions	state=-3	65	65	0	35	0.24298584456930455	65/65
RandomExcursions	state=-2	65	63	2	35	0.26445761931640893	63/65
RandomExcursions	state=-1	65	64	1	35	0.92203553464354604	64/65
RandomExcursions	state=1	65	64	1	35	0.18656597485845114	64/65
RandomExcursions	state=2	65	65	0	35	0.58520876938251232	65/65
RandomExcursions	state=3	65	63	2	35	0.58520876938251232	63/65
RandomExcursions	state=4	65	65	0	35	0.17029390843980358	65/65
RandomExcursionsVariant	state=-9	65	64	1	35	0.55102569688485936	64/65
RandomExcursionsVariant	state=-8	65	64	1	35	0.051390834746201007	64/65
RandomExcursionsVariant	state=-7	65	64	1	35	0.6890190210880669	64/65
RandomExcursionsVariant	state=-6	65	64	1	35	0.55102569688485969	64/65
RandomExcursionsVariant	state=-5	65	64	1	35	0.6890190210880669	64/65
RandomExcursionsVariant	state=-4	65	64	1	35	0.39245576033480623	64/65
RandomExcursionsVariant	state=-3	65	64	1	35	0.17029390843980377	64/65
RandomExcursionsVariant	state=-2	65	63	2	35	0.45279922039701415	63/65
RandomExcursionsVariant	state=-1	65	65	0	35	0.013411357696626415	65/65
RandomExcursionsVariant	state=1	65	65	0	35	0.94114441004072225	65/65
RandomExcursionsVariant	state=2	65	65	0	35	0.58520876938251232	65/65
RandomExcursionsVariant	state=3	65	65	0	35	0.39245576033480623	65/65
RandomExcursionsVariant	state=4	65	65	0	35	0.7564762011090721	65/65
RandomExcursionsVariant	state=5	65	65	0	35	0.58520876938251232	65/65
RandomExcursionsVariant	state=6	65	65	0	35	0.58520876938251254	65/65
RandomExcursionsVariant	state=7	65	65	0	35	0.51744154211758009	65/65
RandomExcursionsVariant	state=8	65	65	0	35	0.45279922039701448	65/65
RandomExcursionsVariant	state=9	65	65	0	35	0.010606135946766487	65/65
Serial	p_value1 (delta_1)	100	98	2	0	0.6993125708664083	98/100
Serial	p_value2 (delta_2)	100	98	2	0	0.97169915710286503	98/100
LinearComplexity	single	100	100	0	0	0.73991829209465376	100/100

Authentication-tag stream

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	99	1	0	0.5141236202310755	99/100
BlockFrequency	single	100	100	0	0	0.57490342386445548	100/100
CumulativeSums	forward	100	100	0	0	0.53414621690969155	100/100
CumulativeSums	reverse	100	98	2	0	0.59554850728420583	98/100
Runs	single	100	99	1	0	0.75975627887206953	99/100
LongestRun	single	100	100	0	0	0.59554850728420594	100/100
Rank	single	100	96	4	0	0.13728235312658429	96/100
FFT	single	100	100	0	0	0.289667490388479	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	99	1	0	0.5749034238644557	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	99	1	0	0.16260626219029947	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	99	1	0	0.38382651737138401	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	99	1	0	0.98345295122950693	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	100	0	0	0.65793330621362878	100/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	99	1	0	0.26224875459496438	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	99	1	0	0.49439168648212251	99/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	99	1	0	0.010988016145239981	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	100	0	0	0.13728235312658429	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	99	1	0	0.071177409391062527	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	100	0	0	0.49439168648212251	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	99	1	0	0.010236918305430562	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	100	0	0	0.63711940716939863	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	99	1	0	0.88317137842416837	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	100	0	0	0.30412581641191022	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	100	0	0	0.35048521232336122	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	97	3	0	0.21330930508341672	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	97	3	0	0.77918771616483629	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	98	2	0	0.63711940716939852	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	99	1	0	0.88317137842416837	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	96	4	0	0.53414621690969155	96/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	99	1	0	0.85138257535679507	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	99	1	0	0.75975627887206953	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	97	3	0	0.65793330621362878	97/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	100	0	0	0.67868578529607482	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	100	0	0	0.22482064155775419	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0	100	100	0	0	0.5749034238644557	100/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	98	2	0	0.096578173851826016	98/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 0	100	99	1	0	0.67868578529607482	99/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	99	1	0	0.67868578529607493	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	99	1	0	0.075719263168633866	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	98	2	0	0.71974657482403437	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	99	1	0	0.6993125708664083	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	98	2	0	0.57490342386445548	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	99	1	0	0.55442043587285705	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	99	1	0	0.36691779911275224	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	98	2	0	0.5141236202310755	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	100	0	0	0.81653679759724718	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	98	2	0	0.21330930508341672	98/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	99	1	0	0.71974657482403392	99/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0	100	100	0	0	0.6163052249833656	100/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1	100	100	0	0	0.5141236202310755	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0	100	99	1	0	0.73991829209465354	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1	100	98	2	0	0.4011992865802575	98/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	100	0	0	0.47498568664800966	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	99	1	0	0.30412581641191022	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	100	0	0	0.93571641091397317	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	100	0	0	0.22482064155775411	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	100	0	0	0.36691779911275224	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	100	0	0	0.57490342386445548	100/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	99	1	0	0.94630767376442704	99/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	100	0	0	0.062820720617879772	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	100	0	0	0.49439168648212251	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	99	1	0	0.63711940716939852	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 0 1 1	100	99	1	0	0.47498568664800966	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	99	1	0	0.75975627887206965	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 0 1 1	100	99	1	0	0.98789554017452808	99/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	100	0	0	0.47498568664800966	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 0 1 1	100	100	0	0	0.002374441119716499	100/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	100	0	0	0.65793330621362922	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 0 1 1	100	98	2	0	0.73991829209465365	98/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	100	0	0	0.35048521232336122	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	99	1	0	0.023545032770843879	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	100	0	0	0.24928389397664924	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	100	0	0	0.13728235312658446	100/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	95	5	0	0.55442043587285705	95/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	99	1	0	0.92407593012659839	99/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	100	0	0	0.11538658258886129	100/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	99	1	0	0.8676916714977051	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	99	1	0	0.26224875459496472	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	99	1	0	0.26224875459496472	99/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	100	0	0	0.61630522498336537	100/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	99	1	0	0.63711940716939852	99/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	99	1	0	0.55442043587285705	99/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	99	1	0	0.5749034238644557	99/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	99	1	0	0.31908350214261383	99/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	99	1	0	0.71974657482403437	99/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	100	0	0	0.43727418891386738	100/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	98	2	0	0.55442043587285705	98/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	99	1	0	0.15376333779556459	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	100	0	0	0.71974657482403392	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 1 0 0	100	99	1	0	0.14532611574989807	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	98	2	0	0.53414621690969133	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	100	0	0	0.91141252683167917	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	100	0	0	0.14532611574989807	100/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	99	1	0	0.36691779911275224	99/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	98	2	0	0.33453815163400619	98/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	100	0	0	0.090935976579805164	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	100	0	0	0.13728235312658429	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	100	0	0	0.4011992865802575	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	97	3	0	0.47498568664800966	97/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	97	3	0	0.40119928658025733	97/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	98	2	0	0.71974657482403437	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	98	2	0	0.85138257535679507	98/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	100	0	0	0.8165367975972474	100/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	97	3	0	0.8165367975972474	97/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	99	1	0	0.63711940716939852	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 1 0	100	99	1	0	0.071177409391062527	99/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	100	0	0	0.040108279765939524	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	99	1	0	0.9914676066288135	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	99	1	0	0.63711940716939852	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	100	0	0	0.98789554017452808	100/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	98	2	0	0.55442043587285705	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	99	1	0	0.6993125708664083	99/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	98	2	0	0.38382651737138401	98/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	100	0	0	0.048715976147641296	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	99	1	0	0.36691779911275224	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	99	1	0	0.65793330621362878	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	99	1	0	0.7791877161648364	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	99	1	0	0.77918771616483629	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	98	2	0	0.75975627887206942	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	100	0	0	0.93571641091397317	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	99	1	0	0.080518512818763724	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	99	1	0	0.2022676854058254	99/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	98	2	0	0.41902116661019273	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	98	2	0	0.5141236202310755	98/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	100	0	0	0.53414621690969155	100/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	99	1	0	0.16260626219029947	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	99	1	0	0.85138257535679507	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	100	0	0	0.83430826019340754	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	98	2	0	0.13728235312658429	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	100	0	0	0.59554850728420583	100/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	99	1	0	0.23680981521773165	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	98	2	0	0.75975627887206942	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	98	2	0	0.21330930508341672	98/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	97	3	0	0.75975627887206942	97/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	99	1	0	0.59554850728420594	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	99	1	0	0.38382651737138435	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	99	1	0	0.289667490388479	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	99	1	0	0.20226768540582532	99/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	97	3	0	0.35048521232336122	97/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	100	0	0	0.81653679759724718	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	100	0	0	0.88317137842416837	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	100	0	0	0.21330930508341639	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	100	0	0	0.30412581641191033	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	100	0	0	0.73991829209465365	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	99	1	0	0.41902116661019245	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	100	0	0	0.21330930508341639	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	97	3	0	0.73991829209465376	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	99	1	0	0.47498568664800989	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	99	1	0	0.6993125708664083	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	100	0	0	0.18155663198089653	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	99	1	0	0.35048521232336122	99/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 1 0	100	100	0	0	0.051941639795524172	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0 0	100	100	0	0	0.36691779911275224	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 1 0	100	97	3	0	0.8165367975972474	97/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0 0	100	100	0	0	0.12961959133276191	100/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 1 0	100	99	1	0	0.55442043587285705	99/100
OverlappingTemplate	single	100	98	2	0	0.10879095519400762	98/100
Universal	single	100	99	1	0	0.61630522498336548	99/100
ApproximateEntropy	single	100	98	2	0	0.92407593012659839	98/100
RandomExcursions	state=-4	62	60	2	38	0.53414621690969155	60/62

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
RandomExcursions	state=-3	62	57	5	38	0.028181144804378212	57/62
RandomExcursions	state=-2	62	62	0	38	0.70614871255331613	62/62
RandomExcursions	state=-1	62	62	0	38	0.83430826019340754	62/62
RandomExcursions	state=1	62	61	1	38	0.27570893677222202	61/62
RandomExcursions	state=2	62	62	0	38	0.67177902289518643	62/62
RandomExcursions	state=3	62	62	0	38	0.40709083549058772	62/62
RandomExcursions	state=4	62	62	0	38	0.96429497268508912	62/62
RandomExcursionsVariant	state=-9	62	61	1	38	0.6717790228951861	61/62
RandomExcursionsVariant	state=-8	62	61	1	38	0.86234438441100869	61/62
RandomExcursionsVariant	state=-7	62	60	2	38	0.83430826019340754	60/62
RandomExcursionsVariant	state=-6	62	61	1	38	0.73991829209465365	61/62
RandomExcursionsVariant	state=-5	62	61	1	38	0.25355072973231113	61/62
RandomExcursionsVariant	state=-4	62	62	0	38	0.53414621690969133	62/62
RandomExcursionsVariant	state=-3	62	60	2	38	0.11095216762055737	60/62
RandomExcursionsVariant	state=-2	62	60	2	38	0.50093444774910523	60/62
RandomExcursionsVariant	state=-1	62	60	2	38	0.054199121015039847	60/62
RandomExcursionsVariant	state=1	62	62	0	38	0.88813658797344031	62/62
RandomExcursionsVariant	state=2	62	62	0	38	0.63711940716939863	62/62
RandomExcursionsVariant	state=3	62	62	0	38	0.054199121015039847	62/62
RandomExcursionsVariant	state=4	62	62	0	38	0.43727418891386727	62/62
RandomExcursionsVariant	state=5	62	62	0	38	0.35048521232336122	62/62
RandomExcursionsVariant	state=6	62	62	0	38	0.43727418891386738	62/62
RandomExcursionsVariant	state=7	62	62	0	38	0.7061487125533159	62/62
RandomExcursionsVariant	state=8	62	62	0	38	0.60245760913961299	62/62
RandomExcursionsVariant	state=9	62	62	0	38	0.9914676066288135	62/62
Serial	p_value1 (delta_1)	100	97	3	0	0.9780720677116449	97/100
Serial	p_value2 (delta_2)	100	100	0	0	0.65793330621362922	100/100
LinearComplexity	single	100	99	1	0	0.6993125708664083	99/100

Encrypted Step-2 image-output diagnostic

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	0	100	0	6.1868010323945917e-188	0/100
BlockFrequency	single	100	0	100	0	6.1868010323945917e-188	0/100
CumulativeSums	forward	100	0	100	0	6.1868010323945917e-188	0/100
CumulativeSums	reverse	100	0	100	0	6.1868010323945917e-188	0/100
Runs	single	100	0	100	0	6.1868010323945917e-188	0/100
LongestRun	single	100	0	100	0	6.1868010323945917e-188	0/100
Rank	single	100	0	100	0	6.1868010323945917e-188	0/100
FFT	single	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	8	92	0	2.6515577458459591e-167	8/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	8	92	0	1.9018599173356807e-179	8/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	11	89	0	3.535205000106014e-171	11/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	18	82	0	3.3306752060725173e-136	18/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	15	85	0	4.4130757532546044e-159	15/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	7	93	0	1.9018599173356807e-179	7/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	9	91	0	2.6515577458459591e-167	9/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	29	71	0	3.3329891580383414e-122	29/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	10	90	0	2.1283956007471158e-175	10/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	44	56	0	1.2278418979728443e-92	44/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	14	86	0	2.8993004069920525e-171	14/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	2	98	0	1.1408899415220089e-183	2/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	3	97	0	6.1868010323945917e-188	3/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	11	89	0	2.8658897638857205e-175	11/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	36	64	0	8.8585899019420294e-106	36/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	33	67	0	6.9551354520108655e-99	33/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	29	71	0	8.2396786369415659e-116	29/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	17	83	0	7.4431257671925874e-144	17/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0	100	14	86	0	3.535205000106014e-171	14/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	27	73	0	1.3541018035536664e-125	27/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	5	95	0	1.1408899415220089e-183	5/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	48	52	0	3.7663979731142584e-93	48/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	11	89	0	2.444948253582859e-155	11/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	24	76	0	2.25842401860276e-140	24/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	5	95	0	6.1868010323945917e-188	5/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	8	92	0	1.1408899415220089e-183	8/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0	100	20	80	0	2.9807004406678982e-155	20/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0	100	23	77	0	5.5036091452119337e-140	23/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1	100	45	55	0	1.4871738213146354e-73	45/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	2	98	0	6.1868010323945917e-188	2/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	2	98	0	6.1868010323945917e-188	2/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	5	95	0	2.1283956007471158e-175	5/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	9	91	0	1.1408899415220089e-183	9/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	10	90	0	2.1283956007471158e-175	10/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 0 1 1	100	52	48	0	2.9203825879544675e-84	52/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 0 1 1	100	26	74	0	2.4524688021470808e-112	26/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	25	75	0	2.008526258681169e-147	25/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 0 1 1	100	21	79	0	1.1088020447559905e-147	21/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 0 1 1	100	17	83	0	3.9967567520191655e-159	17/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	19	81	0	4.9717646302841826e-129	19/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	25	75	0	1.8215841187544815e-125	25/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	2	98	0	1.1408899415220089e-183	2/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	4	96	0	1.9018599173356807e-179	4/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	21	79	0	6.1353826774257756e-133	21/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	26	74	0	3.0323616435765152e-129	26/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	8	92	0	2.6515577458459591e-167	8/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	1	99	0	1.1408899415220089e-183	1/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	50	50	0	3.7975547364173797e-68	50/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	8	92	0	1.7222708924704137e-179	8/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 1 0 0	100	15	85	0	1.4923349223409742e-147	15/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	4	96	0	6.1868010323945917e-188	4/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	18	82	0	1.3516440173772289e-147	18/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	36	64	0	1.2149862550997909e-118	36/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	7	93	0	3.9036885580523575e-171	7/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	9	91	0	2.5953176714988682e-175	9/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	6	94	0	2.1283956007471158e-175	6/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	19	81	0	3.9731375543327961e-163	19/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	45	55	0	3.6454734937487553e-106	45/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	55	45	0	9.0518904526731557e-49	55/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	38	62	0	1.3562417382334614e-112	38/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	31	69	0	1.4367360967432702e-99	31/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	3	97	0	6.1868010323945917e-188	3/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	13	87	0	4.4130757532546044e-159	13/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	11	89	0	3.6338463458609125e-155	11/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 1 0	100	13	87	0	3.6197114299724116e-159	13/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	8	92	0	2.5953176714988682e-175	8/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	7	93	0	2.1283956007471158e-175	7/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	16	84	0	6.6887729171356289e-129	16/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	12	88	0	3.9419382410454e-167	12/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	36	64	0	1.9524170385347919e-108	36/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	13	87	0	3.7044102406122881e-140	13/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	6	94	0	6.1868010323945917e-188	6/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	32	68	0	3.3329891580381495e-122	32/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	35	65	0	9.7849731374217583e-109	35/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	2	98	0	6.1868010323945917e-188	2/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	31	69	0	5.9739043368934561e-109	31/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	14	86	0	4.844171915706041e-163	14/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	16	84	0	1.8215841187544815e-125	16/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	9	91	0	2.5953176714988682e-175	9/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	15	85	0	8.4470081602586011e-149	15/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	12	88	0	3.535205000106014e-171	12/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	5	95	0	1.9018599173356807e-179	5/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	43	57	0	5.8124769809123598e-84	43/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	4	96	0	6.1868010323945917e-188	4/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	37	63	0	1.3099004942943096e-102	37/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	25	75	0	5.5575378619345311e-133	25/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	4	96	0	1.1408899415220089e-183	4/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	45	55	0	7.2418081637386132e-87	45/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	43	57	0	5.0609745073220358e-93	43/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	6	94	0	6.1868010323945917e-188	6/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	10	90	0	2.8993004069920525e-171	10/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	11	89	0	4.844171915706041e-163	11/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	12	88	0	6.1868010323945917e-188	12/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	32	68	0	6.6923902967998439e-90	32/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0 0	100	0	100	0	6.1868010323945917e-188	0/100
OverlappingTemplate	single	100	0	100	0	6.1868010323945917e-188	0/100
Universal	single	100	0	100	0	6.1868010323945917e-188	0/100
ApproximateEntropy	single	100	0	100	0	6.1868010323945917e-188	0/100
RandomExcursions	state=-4	0	0	0	100	----	-----

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
RandomExcursions	state=-3	0	0	0	100	----	-----
RandomExcursions	state=-2	0	0	0	100	----	-----
RandomExcursions	state=-1	0	0	0	100	----	-----
RandomExcursions	state=1	0	0	0	100	----	-----
RandomExcursions	state=2	0	0	0	100	----	-----
RandomExcursions	state=3	0	0	0	100	----	-----
RandomExcursions	state=4	0	0	0	100	----	-----
RandomExcursionsVariant	state=-9	0	0	0	100	----	-----
RandomExcursionsVariant	state=-8	0	0	0	100	----	-----
RandomExcursionsVariant	state=-7	0	0	0	100	----	-----
RandomExcursionsVariant	state=-6	0	0	0	100	----	-----
RandomExcursionsVariant	state=-5	0	0	0	100	----	-----
RandomExcursionsVariant	state=-4	0	0	0	100	----	-----
RandomExcursionsVariant	state=-3	0	0	0	100	----	-----
RandomExcursionsVariant	state=-2	0	0	0	100	----	-----
RandomExcursionsVariant	state=-1	0	0	0	100	----	-----
RandomExcursionsVariant	state=1	0	0	0	100	----	-----
RandomExcursionsVariant	state=2	0	0	0	100	----	-----
RandomExcursionsVariant	state=3	0	0	0	100	----	-----
RandomExcursionsVariant	state=4	0	0	0	100	----	-----
RandomExcursionsVariant	state=5	0	0	0	100	----	-----
RandomExcursionsVariant	state=6	0	0	0	100	----	-----
RandomExcursionsVariant	state=7	0	0	0	100	----	-----
RandomExcursionsVariant	state=8	0	0	0	100	----	-----
RandomExcursionsVariant	state=9	0	0	0	100	----	-----
Serial	p_value1 (delta_1)	100	0	100	0	6.1868010323945917e-188	0/100
Serial	p_value2 (delta_2)	100	0	100	0	6.1868010323945917e-188	0/100
LinearComplexity	single	100	97	3	0	0.6163052249833656	97/100

Final encrypted image with embedded authentication information

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
Frequency	single	100	0	100	0	6.1868010323945917e-188	0/100
BlockFrequency	single	100	0	100	0	6.1868010323945917e-188	0/100
CumulativeSums	forward	100	0	100	0	6.1868010323945917e-188	0/100
CumulativeSums	reverse	100	0	100	0	6.1868010323945917e-188	0/100
Runs	single	100	0	100	0	6.1868010323945917e-188	0/100
LongestRun	single	100	0	100	0	6.1868010323945917e-188	0/100
Rank	single	100	0	100	0	6.1868010323945917e-188	0/100
FFT	single	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 0 1	100	4	96	0	1.1408899415220089e-183	4/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 0 1 1	100	9	91	0	2.9512360342810254e-163	9/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 0 1	100	11	89	0	1.9018599173356807e-179	11/100
NonOverlappingTemplate	m=9 template=0 0 0 0 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 0 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 0 1 1	100	18	82	0	6.0761625165163139e-140	18/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 0 1	100	16	84	0	1.1059309960533784e-143	16/100
NonOverlappingTemplate	m=9 template=0 0 0 0 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	10	90	0	2.5953176714988682e-175	10/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	12	88	0	3.535205000106014e-171	12/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	36	64	0	1.8494712712227826e-129	36/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	2	98	0	6.1868010323945917e-188	2/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	36	64	0	1.3329211023811195e-89	36/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	13	87	0	3.2329995191515996e-167	13/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 0 1	100	1	99	0	1.1408899415220089e-183	1/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 0	100	4	96	0	1.9018599173356807e-179	4/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 0 1 1	100	2	98	0	6.1868010323945917e-188	2/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 0	100	17	83	0	2.444948253582859e-155	17/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 0 1	100	5	95	0	6.1868010323945917e-188	5/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 0	100	40	60	0	1.655535861375466e-87	40/100
NonOverlappingTemplate	m=9 template=0 0 0 1 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 0 1	100	26	74	0	2.2122285214210666e-115	26/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 0	100	37	63	0	1.7671417978227482e-105	37/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 0 1 1	100	24	76	0	1.8395434193775047e-136	24/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 0	100	14	86	0	2.5953176714988682e-175	14/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 0 1	100	26	74	0	4.4346837306946443e-112	26/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 0 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 0 1	100	5	95	0	1.9018599173356807e-179	5/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 0	100	2	98	0	1.1408899415220089e-183	2/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 0 1 1	100	46	54	0	5.4964991267882108e-90	46/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 0 1	100	12	88	0	1.3520132619909782e-151	12/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 0	100	19	81	0	2.4752654930680996e-136	19/100
NonOverlappingTemplate	m=9 template=0 0 1 0 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 0 1	100	8	92	0	1.1408899415220089e-183	8/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 0 1 1	100	10	90	0	1.7222708924704137e-179	10/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 0 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 0	100	21	79	0	3.291110769952678e-155	21/100
NonOverlappingTemplate	m=9 template=0 0 1 0 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 0	100	25	75	0	4.5036533483946593e-129	25/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 0 1 1	100	44	56	0	2.2018145257734292e-73	44/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 0 1 1	100	2	98	0	6.1868010323945917e-188	2/100
NonOverlappingTemplate	m=9 template=0 0 1 1 0 1 1 1 1	100	11	89	0	3.2329995191515996e-167	11/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 0 1	100	5	95	0	1.9018599173356807e-179	5/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 0 1 1 1	100	18	82	0	4.387090802864172e-163	18/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 0 1 1	100	17	83	0	1.3520132619909782e-151	17/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 0 1	100	2	98	0	6.1868010323945917e-188	2/100
NonOverlappingTemplate	m=9 template=0 0 1 1 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 0 1 1	100	59	41	0	5.2914449180824292e-56	59/100
NonOverlappingTemplate	m=9 template=0 1 0 0 0 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 0 1 1	100	26	74	0	1.34114708187481e-118	26/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 0 1 1 1	100	24	76	0	4.1347103073635833e-126	24/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 0 1 1	100	19	81	0	8.2383580660391797e-148	19/100
NonOverlappingTemplate	m=9 template=0 1 0 0 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 0 1 1	100	23	77	0	9.0958907226184508e-148	23/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 0 1 1 1	100	26	74	0	8.1836939827498564e-119	26/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 0 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 0 1 1 1 1	100	23	77	0	3.3475511007323001e-129	23/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 0 1 1	100	6	94	0	1.1408899415220089e-183	6/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 0 1 1 1	100	6	94	0	1.9018599173356807e-179	6/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 0 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 0 1 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 0 1 1 1	100	24	76	0	8.260091960590233e-126	24/100
NonOverlappingTemplate	m=9 template=0 1 1 0 0 1 1 1 1	100	27	73	0	2.0108586544177911e-125	27/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 0 1 1 1	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=0 1 1 0 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 0 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 0 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=0 1 1 1 1 1 1 1 1	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 0 0 0 0 0 0 0 0	100	4	96	0	1.1408899415220089e-183	4/100
NonOverlappingTemplate	m=9 template=1 0 0 0 1 0 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 0 0 0 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 0 0 1 0 1 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 0 0 0 0	100	1	99	0	1.1408899415220089e-183	1/100
NonOverlappingTemplate	m=9 template=1 0 0 1 1 1 0 0 0	100	51	49	0	1.5473137133164916e-55	51/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 0 0 0	100	5	95	0	1.7222708924704137e-179	5/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 0 1 0 0	100	13	87	0	3.6338463458609125e-155	13/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 0 0 0	100	6	94	0	6.1868010323945917e-188	6/100
NonOverlappingTemplate	m=9 template=1 0 1 0 0 1 1 0 0	100	6	94	0	1.7222708924704137e-179	6/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 0 0 0	100	32	68	0	4.9717646302841826e-129	32/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 0 1 0 0	100	8	92	0	2.5953176714988682e-175	8/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 0 0 0	100	14	86	0	3.535205000106014e-171	14/100
NonOverlappingTemplate	m=9 template=1 0 1 0 1 1 1 0 0	100	7	93	0	1.1408899415220089e-183	7/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 0 0 0	100	17	83	0	4.8063303412163879e-167	17/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 0 1 0 0	100	35	65	0	4.1474655325135566e-113	35/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 0 0 0	100	57	43	0	6.477328131965843e-73	57/100
NonOverlappingTemplate	m=9 template=1 0 1 1 0 1 1 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 0 0 0	100	40	60	0	4.4429853931017005e-109	40/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 0 1 0 0	100	33	67	0	9.3480804468785538e-99	33/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 0 0 0	100	15	85	0	3.2015033042094182e-171	15/100
NonOverlappingTemplate	m=9 template=1 0 1 1 1 1 1 0 0	100	10	90	0	1.9018599173356807e-179	10/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 0 0	100	12	88	0	4.4130757532546044e-159	12/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 0 1 0	100	13	87	0	3.9967567520191655e-159	13/100

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
NonOverlappingTemplate	m=9 template=1 1 0 0 0 0 1 0 0	100	5	95	0	1.7222708924704137e-179	5/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 0 0	100	10	90	0	2.6515577458459591e-167	10/100
NonOverlappingTemplate	m=9 template=1 1 0 0 0 1 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 0 0	100	18	82	0	2.2689522163215774e-144	18/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 0 1 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 0 1 0 0	100	15	85	0	3.9967567520191655e-159	15/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 0 0	100	33	67	0	2.6123105952016915e-102	33/100
NonOverlappingTemplate	m=9 template=1 1 0 0 1 1 0 1 0	100	16	84	0	5.5036091452122516e-140	16/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 0 0	100	7	93	0	1.9018599173356807e-179	7/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 0 1 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 0 1 0 0	100	32	68	0	1.8038086073871188e-118	32/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 0 0	100	39	61	0	4.2086774224063045e-96	39/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 0 1 0 1 1 0 0	100	4	96	0	6.1868010323945917e-188	4/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 0 0	100	30	70	0	5.9739043368934561e-109	30/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 0 1 0	100	14	86	0	3.6338463458609125e-155	14/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 0 1 0 0	100	16	84	0	4.079615146817471e-129	16/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 0 0	100	10	90	0	2.8993004069920525e-171	10/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 0 1 0	100	16	84	0	4.1095341827450426e-144	16/100
NonOverlappingTemplate	m=9 template=1 1 0 1 1 1 1 0 0	100	13	87	0	1.9974190138214381e-159	13/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 0 0	100	3	97	0	1.1408899415220089e-183	3/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 0 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 0 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 0 1 1 0	100	40	60	0	4.7748182010698369e-84	40/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 0 0	100	5	95	0	1.1408899415220089e-183	5/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 0 1 0	100	42	58	0	7.2650690323064558e-112	42/100
NonOverlappingTemplate	m=9 template=1 1 1 0 0 1 1 0 0	100	29	71	0	2.4752654930682405e-136	29/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 0 0	100	3	97	0	6.1868010323945917e-188	3/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 0 1 0	100	49	51	0	1.2954691648420979e-81	49/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 0 0	100	42	58	0	4.2129540133097677e-81	42/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 0 1 1 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 0 0	100	5	95	0	1.7222708924704137e-179	5/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 0 1 0	100	9	91	0	3.9419382410454e-167	9/100
NonOverlappingTemplate	m=9 template=1 1 1 0 1 1 1 0 0	100	15	85	0	6.0761625165159671e-140	15/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 0 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 0 1 0	100	3	97	0	1.1408899415220089e-183	3/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 0 1 1 0	100	16	84	0	2.9512360342810254e-163	16/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 0 1 0	100	32	68	0	4.9812479037048594e-90	32/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 0 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 0 1 1 1 0	100	4	96	0	6.1868010323945917e-188	4/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 0 1 0	100	2	98	0	6.1868010323945917e-188	2/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 0 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 0 1 1 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 0 1 0	100	1	99	0	6.1868010323945917e-188	1/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 0 0	100	0	100	0	6.1868010323945917e-188	0/100
NonOverlappingTemplate	m=9 template=1 1 1 1 1 1 1 1 0	100	0	100	0	6.1868010323945917e-188	0/100
OverlappingTemplate	single	100	0	100	0	6.1868010323945917e-188	0/100
Universal	single	100	0	100	0	6.1868010323945917e-188	0/100
ApproximateEntropy	single	100	0	100	0	6.1868010323945917e-188	0/100
RandomExcursions	state=-4	0	0	0	100	----	-----

NIST test	Component	Valid	Pass	Fail	N/A	Uniformity p-value	Pass proportion
RandomExcursions	state=-3	0	0	0	100	----	-----
RandomExcursions	state=-2	0	0	0	100	----	-----
RandomExcursions	state=-1	0	0	0	100	----	-----
RandomExcursions	state=1	0	0	0	100	----	-----
RandomExcursions	state=2	0	0	0	100	----	-----
RandomExcursions	state=3	0	0	0	100	----	-----
RandomExcursions	state=4	0	0	0	100	----	-----
RandomExcursionsVariant	state=-9	0	0	0	100	----	-----
RandomExcursionsVariant	state=-8	0	0	0	100	----	-----
RandomExcursionsVariant	state=-7	0	0	0	100	----	-----
RandomExcursionsVariant	state=-6	0	0	0	100	----	-----
RandomExcursionsVariant	state=-5	0	0	0	100	----	-----
RandomExcursionsVariant	state=-4	0	0	0	100	----	-----
RandomExcursionsVariant	state=-3	0	0	0	100	----	-----
RandomExcursionsVariant	state=-2	0	0	0	100	----	-----
RandomExcursionsVariant	state=-1	0	0	0	100	----	-----
RandomExcursionsVariant	state=1	0	0	0	100	----	-----
RandomExcursionsVariant	state=2	0	0	0	100	----	-----
RandomExcursionsVariant	state=3	0	0	0	100	----	-----
RandomExcursionsVariant	state=4	0	0	0	100	----	-----
RandomExcursionsVariant	state=5	0	0	0	100	----	-----
RandomExcursionsVariant	state=6	0	0	0	100	----	-----
RandomExcursionsVariant	state=7	0	0	0	100	----	-----
RandomExcursionsVariant	state=8	0	0	0	100	----	-----
RandomExcursionsVariant	state=9	0	0	0	100	----	-----
Serial	p_value1 (delta_1)	100	0	100	0	6.1868010323945917e-188	0/100
Serial	p_value2 (delta_2)	100	0	100	0	6.1868010323945917e-188	0/100
LinearComplexity	single	100	99	1	0	0.41902116661019273	99/100

Interpreting very small first-level p-values

The following are the 12 smallest applicable first-level values in the complete run, shown to explain the numeric edge cases; this is not a filtered result set. Every stream/test/component result, including all failures, is retained in results.csv and `p_values_full_precision.csv.

Category	Stream	Test	Component	First-level p-value	Status
Encrypted Step-2 image-output diagnostic	1	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	2	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	3	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	4	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	6	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	7	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	8	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	9	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	10	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	12	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	13	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow
Encrypted Step-2 image-output diagnostic	14	Frequency	single	<1e-300 (STS binary64 underflow)	double_underflow

Individual p-values and official reports

- results.csv and p_values_full_precision.csv contain one row for every stream and test component, the full first-level value, a zero/N/A status, any underflow bound, the uniformity p-value, alpha, bit length, and raw input stream hash.

- `authentication_tag_sample_digests.csv` provides SHA-256 digests of one actual raw 256-bit HMAC tag from each of 100 independent key/ImageID cases; no secret key, ImageID, or raw tag bytes are published.
- `per_test_summary.csv` contains the complete per-test/component summary.
- `raw_reports/` preserves the complete high-precision rerun, including `finalAnalysisReport.txt`, `per-test stats.txt`, `master results.txt`, and component partitions. `official_native_6dp/` preserves the six original unmodified native `finalAnalysisReport.txt` files.
- The three high-precision category reports are `authentication_tag_stream_finalAnalysisReport.txt`, `encrypted_step2_finalAnalysisReport.txt`, and `final_marked_authenticated_finalAnalysisReport.txt`. Their official native counterparts are `official_authentication_tag_stream_finalAnalysisReport.txt`, `official_encrypted_step2_finalAnalysisReport.txt`, and `official_final_marked_authenticated_finalAnalysisReport.txt`.
- `raw_streams/` contains the exact packed binary NIST inputs. `stream_manifest.csv` records each stream's byte offset, size, hash, key/ImageID digests and source details.

Provenance and reproducibility

See `provenance.txt`, `stream_manifest.csv`, `authentication_tag_stream_manifest.csv`, `authentication_tag_sample_digests.csv`, `component_contexts.csv`, and `nist_sts_source_manifest.csv`. They record source and executable hashes, runtime versions, platform, exact command and settings, image/key/ImageID/tag/stream digests, and test credential identifiers. No key or ImageID bytes are written.

Limitations

- NIST SP 800-22 is not a security proof and does not replace cryptanalysis.
- Image-output streams preserve image-related information by design; NIST results on those streams are diagnostics, not proof of security or insecurity.
- NPCR/UACI and NIST outcomes do not prove cryptographic security.
- The evaluation uses six public UCT images because the eight CelebA-HQ source files were unavailable. Authenticated-TPE is implemented with a fixed 32×32 block size; authenticated 8×8 and 16×16 results are therefore not reported.
- The component fixtures are deterministic and public for reproducibility. NIST testing does not assess entropy quality, key secrecy, unpredictability, formal security, or independent cryptanalysis.

References

1. NIST, SP 800-22 Rev. 1a. NIST notes statistical testing cannot substitute for cryptanalysis.
2. NIST, SP 800-22 documentation and software, including the NIST STS download and software revision history.